Zhe Xu

Research Scientist @ Meta

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Key Expertise

LLM & LLM-native RecSys; Generative Modeling; Graph Machine Learning.

Education

2021–2025 **Ph.D. in Computer Science**, *University of Illinois Urbana-Champaign*
Thesis: *Optimal Graph Learning*.

2019–2021 **M.S. in Computer Science**, *University of Illinois Urbana-Champaign*
Thesis: *Dense Subgraph Detection on Multi-layered Networks*.

2014–2018 **B.E. in Electronic Engineering**, *Fudan University*, China

Publications

Conference Papers

C1 **Zhe Xu**, Ankit Peshin, Chiyu Zhang, Feng Qi, Johnson Lui, Anil Ramakrishna, et al. *Tokenizing Numerical and Embedding Features for LLM RecSys*. TokShop at COLM 2026.

C2 **Zhe Xu**, Kaveh Hassani, Si Zhang, Hanqing Zeng, Michihiro Yasunaga, Limei Wang, Dongqi Fu, Ning Yao, Bo Long, Hanghang Tong. *Language Models are Graph Learners*. Findings of EACL 2026.

C3 **Zhe Xu**, Menghai Pan, Yuzhong Chen, Huiyuan Chen, Yuchen Yan, Mahashweta Das, Hanghang Tong. *Fine-grained Graph Rationalization*. CIKM 2025.

C4 **Zhe Xu**, Ruizhong Qiu, Yuzhong Chen, Huiyuan Chen, Xiran Fan, Menghai Pan, Zhichen Zeng, Mahashweta Das, Hanghang Tong. *Discrete-state Continuous-time Diffusion for Graph Generation*. NeurIPS 2024.

C5 **Zhe Xu**, Yuzhong Chen, Menghai Pan, Huiyuan Chen, Mahashweta Das, Hao Yang, Hanghang Tong. *Kernel Ridge Regression-Based Graph Dataset Distillation*. SIGKDD 2023.

C6 **Zhe Xu**, Yuzhong Chen, Qinghai Zhou, Yuhang Wu, Menghai Pan, Hao Yang, Hanghang Tong. *Node Classification Beyond Homophily: Towards a General Solution*. SIGKDD 2023.

C7 **Zhe Xu**, Kaize Ding, Yu-Xiong Wang, Huan Liu, Hanghang Tong. *Generalized Few-Shot Node Classification on Graphs*. ICDM 2022 (**Best-Ranked Papers**).

C8 **Zhe Xu**, Boxin Du, Hanghang Tong. *Graph Sanitation with Application to Node Classification*. TheWebConf 2022.

C9 **Zhe Xu**, Si Zhang, Yinglong Xia, Liang Xiong, Jiejun Xu, Hanghang Tong. *DESTINE: Dense Subgraph Detection on Multi-Layered Networks*. CIKM 2021.

C10 **Zhe Xu**, Si Zhang, Yinglong Xia, Liang Xiong, Hanghang Tong. *Ranking on Network of Heterogeneous Information Networks*. IEEE BigData 2020.

C11 Dongqi Fu*, **Zhe Xu***, Bo Li, Hanghang Tong, Jingrui He. *A View-Adversarial Framework for Multi-View Network Embedding*. CIKM 2020. (*equal contribution)

C12 Ruizhong Qiu, **Zhe Xu**, Wenxuan Bao, Hanghang Tong. *Turing Completeness of Prompting*. ICLR 2025.

C13 Huiyuan Chen, **Zhe Xu**, Chin-Chia Michael Yeh, Vivian Lai, Yan Zheng, Minghua Xu, Hanghang Tong. *Masked Graph Transformer for Large-Scale Recommendation*. SIGIR 2024.

C14 Qinghai Zhou, Yuzhong Chen, **Zhe Xu**, Yuhang Wu, Menghai Pan, Mahashweta Das, Hao Yang, Hanghang Tong. *Graph Anomaly Detection with Adaptive Node Mixup*. CIKM 2024.

C15 Zhichen Zeng, Ruizhong Qiu, **Zhe Xu**, Zhining Liu, Yuchen Yan, Tianxin Wei, Lei Ying, Jingrui He, Hanghang Tong. *Graph Mixup on Approximate Gromov–Wasserstein Geodesics*. ICML 2024.

C16 Haiyun Jiang, Li Cui, **Zhe Xu**, Deqing Yang, et al. *Relation Extraction Using Supervision from Topic Knowledge of Relation Labels*. IJCAI 2019.

C17 Dongqi Fu*, Liri Fang*, Zihao Li*, **Zhe Xu***, Hanghang Tong, Vetle I Torvik, Jingrui He. *Bring Complex Geometric Information to LLMs: A Positional Survey of Graph Parametric Representation*. LOG 2025. (*equal contribution)

C18 Haobo Xu, Yuchen Yan, Dingsu Wang, **Zhe Xu**, Zhichen Zeng, Tarek F. Abdelzaher, Jiawei Han, Hanghang Tong. *SLOG: An Inductive Spectral Graph Neural Network Beyond Polynomial Filter*. ICML 2024.

C19 Xinyu He, Chenhan Xiao, Haoran Li, Ruizhong Qiu, **Zhe Xu**, Yang Weng, Jingrui He, Hanghang Tong. *PowerGrow: Feasible Co-Growth of Structures and Dynamics for Power Grid Synthesis*. SIGKDD 2026.

C20 Zhining Liu, Ruizhong Qiu, Zhichen Zeng, Hyunsik Yoo, David Zhou, **Zhe Xu**, Yada Zhu, Kommy Weldemariam, Jingrui He, Hanghang Tong. *Class-Imbalanced Graph Learning without Class Rebalancing*. ICML 2024.

Journal Papers

J1 **Zhe Xu**, Kaize Ding, Yu-Xiong Wang, Huan Liu, Hanghang Tong. *Generalized Few-Shot Node Classification on Graphs: Toward an Uncertainty-Based Solution*. KAIS 2023.

J2 Kaize Ding, **Zhe Xu**, Hanghang Tong, Huan Liu. *Data Augmentation for Deep Graph Learning: A Survey*. SIGKDD Explorations 2022.

J3 Qing Chen, Nan Chen, Guande Wu, Ziyan Liu, **Zhe Xu**, Hanghang Tong, Nan Cao. *Calliope-Net: Automatic Generation of Graph Data Facts via Annotated Node-link Diagrams*. IEEE Transactions on Visualization and Computer Graphics 2023.

J4 Zihao Li, Zhichen Zeng, Xiao Lin, Feihao Fang, Yanru Qu, **Zhe Xu**, Zhining Liu, Xuying Ning, Tianxin Wei, Ge Liu, Hanghang Tong, Jingrui He. *Flow Matching Meets Biology and Life Science: A Survey*. npj Artificial Intelligence 2025.

Preprints

P1 **Zhe Xu**, et al. *Card-Shuffling Diffusion Generative Reranker*. Under review.

P2 **Zhe Xu**, et al. *The Case Against Generation for Retrieval: Discriminative Language Models as Effective Retrievers*. Under review.

P3 Xinyu Zhu, **Zhe Xu**, Xiaohan Wei, Yunchen Pu, Fei Tian, Chonglin Sun, Kaushik Rangadurai, et al. *Self-Guided Test-Time Training for Long-Context LLMs*. Under review.

P4 Yufei Li, Zaiwei Zhang, Mingfu Liang, Kavosh Asadi, **Jay Xu**, Jimmy Kim, Chongyang Bai, Jieyi Zhang, et al. *GR2 Technical Report*. Under review.

P5 Mingfu Liang*, Yufei Li*, **Jay Xu***, Kavosh Asadi, Xi Liu, et al. *Generative Reasoning Re-ranker*. Under review.

P6 Beidi Zhao, Boxin Du, **Zhe Xu**, Liangyue Li, Hanghang Tong. *Learning Optimal Propagation for Graph Neural Networks*. Under review.

P7 Yuchen Yan, Yuzhong Chen, Huiyuan Chen, Xiaoting Li, **Zhe Xu**, Mahashweta Das, Hanghang Tong. *THiGCN: Temporal Heterophilic Graph Convolutional Network for Event-based Continuous Graphs*. Under review.

Work Experience

2025–Present **Research Scientist**, *Meta, Monetization*, Sunnyvale, CA
LLM-native ads recommender systems

2024 **Research Intern**, *Meta, Monetization*, Sunnyvale, CA
LLM & graph machine learning

2023 **Ph.D. Intern**, *Visa Research*, Palo Alto, CA
Graph data augmentation

2022 **Research Intern**, *Meta, Applied Machine Learning*, New York, NY
Large-scale GNN systems on organic recommender systems

2019–2025 **Graduate Research Assistant**, *UIUC*, Urbana, IL

2018–2019 **Graduate Research Assistant**, *Arizona State University*, Tempe, AZ

2016–2018 **Undergraduate Research Assistant**, *Fudan University*, Shanghai, China

Talks

2026 *Generative Reasoning Reranker*. Invited Talk, NUS AI Institute.

2023 *Natural and Artificial Dynamics in GNNs: A Tutorial*. Lecture-Style Tutorial, WSDM.

2023 *Graph Data Distillation*. Invited Talk, Visa Research.

Grant Experience

NSF SLES: NetSafe — Safe Graph Neural Networks.

NSF Collaborative Research — Optimal Deep Graph Learning.

Teaching Experience

2023 **Teaching Assistant**, *CS 514 Advanced Topics in Network Science*, UIUC

2021–2022 **Teaching Assistant**, *CS 512 Data Mining Principles*, UIUC

2021 **Teaching Assistant**, *CS 412 Introduction to Data Mining*, UIUC

Service

Conference Reviewer AISTATS, ICML, SDM, ICLR, AAAI, NeurIPS, KDD, CIKM, WWW.

Journal Reviewer TNNLS, TVCG, TKDD.

Workshop Reviewer GLFrontiers, TrustLOG.